Provability facts for the conditional in the sequent calculus

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the census lists prop:provability-lif-left and prop:provability-lif-right on the same statement/proof span; they are sub-labels of this proposition and are covered by this card (same-span consolidation precedent)

Editorial note. The label prop:provability-lif names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Left and right conditional rules. Left-if derives (A implies B) on the left from B on the left and A on the right; Right-if derives (A implies B) on the right from A on the left and B on the right.. Item (1) uses Left-if on the initial sequents; item (2) chains Left-not, Left-exchange, Right-weakening and Right-if in the first tree, and Left-weakening plus Right-if in the second.

Proposition 0.1. *1.* $A, A \to B \vdash B$.

2. Both $\neg A \vdash A \to B$ *and* $B \vdash A \to B$.

Proof. 1. The sequent $A \to B, A \Rightarrow B$ is derivable:

$$\frac{AA \qquad BB}{A \to B, AB}$$

2. Both sequents $\neg A \Rightarrow A \to B$ and $B \Rightarrow A \to B$ are derivable:

$$\dfrac{\dfrac{\dfrac{\dfrac{AA}{\neg A, A}}{A, \neg A}}{A, \neg AB}}{\neg AA \to B} \qquad \dfrac{\dfrac{BB}{A, BB}}{BA \to B}$$

□